

Boston House Price Prediction Using Linear Regression

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1. Introduction

This project predicts house prices using the Boston Housing Dataset and Linear Regression.

2. Problem Statement

The goal is to predict the median house price (MEDV) using different housing features.

3. Dataset Features

CRIM - Crime rate

ZN - Residential land zone

INDUS - Industrial area proportion

CHAS - Charles River dummy variable

NOX - Nitric oxide concentration

RM - Average rooms

AGE - Age of houses

DIS - Distance to employment centers

RAD - Highway accessibility

TAX - Property tax rate

PTRATIO - Student-teacher ratio

B - Population statistic

LSTAT - Lower status population percentage

MEDV - Median house value

4. Technologies Used

- Python

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

5. Workflow

- Load Dataset
- Data Cleaning
- EDA
- Train-Test Split
- Model Training
- Prediction
- Evaluation
- Visualization

6. Data Preprocessing

Missing values are handled using mean imputation.

7. Exploratory Data Analysis

EDA includes:

- Correlation Heatmap
- Histograms
- Scatterplots
- Boxplots

8. Linear Regression

Linear Regression predicts continuous values using:

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

9. Model Evaluation

Metrics used:

- MAE
- MSE
- RMSE
- R2 Score

10. Conclusion

The project successfully predicts house prices using Linear Regression.